```
#include<iostream>
#include<cmath>
#include<cstdlib>
#include<cctype>
using namespace std;
void main()
{
cout<<"abs(3.5):     "<<abs(3.5)<<endl;
cout<<"abs(-3.5):    "<<abs(-3.5)<<endl;
cout<<"ceil(59.76):  "<<ceil(59.76)<<endl;
cout<<"ceil(-59.76): "<<ceil(-59.76)<<endl;
cout<<"exp(1) :      "<<exp(1)<<endl;
cout<<"fabs(3.5):    "<<fabs(3.5)<<endl;
cout<<"cos(0):       "<<cos(0)<<endl;
cout<<"floor(40.8):  "<<floor(40.8)<<endl;
cout<<"floor(-40.8): "<<floor(-40.8)<<endl;
cout<<"tolower(65):  "<<tolower(65)<<endl;
cout<<"toupper(97) : "<<toupper(97)<<endl;

}
```

File Edit View Insert Project Build Tools Window Help

bals) (All global members) main

mmmmmm classes

```
#include<iostream>
#include<cmath>
#include<cstdlib>
#include<cctype>
using namespace std;
void main()
{
cout<<"abs(3.5):      "<<abs(3.5)<<endl;
cout<<"abs(-3.5):     "<<abs(-3.5)<<endl;
cout<<"ceil(59.76):   "<<ceil(59.76)<<endl;
cout<<"ceil(-59.76):  "<<ceil(-59.76)<<endl;
cout<<"exp(1) :       "<<exp(1)<<endl;
cout<<"fabs(3.5):     "<<fabs(3.5)<<endl;
cout<<"cos(0):        "<<cos(0)<<endl;
cout<<"floor(40.8):   "<<floor(40.8)<<endl;
cout<<"floor(-40.8):  "<<floor(-40.8)<<endl;
cout<<"tolower(65):   "<<tolower(65)<<endl;
cout<<"toupper(97) :  "<<toupper(97)<<endl;

}
```

"C:\Program Files\Microsoft Visual Stud...

```
abs(3.5):     3
abs(-3.5):    3
ceil(59.76):  60
ceil(-59.76): -59
exp(1) :      2.71828
fabs(3.5):    3.5
cos(0):       1
floor(40.8):  40
floor(-40.8): -41
tolower(65):  97
toupper(97) : 65
Press any key to continue
```

ClassView FileView

```
inking...

mmmmmm.exe - 0 error(s), 0 warning(s)
```

Build Debug Find in Files 1 Find in Files 2 Results SQL Debugging

Ln 1, Col 1 REC COL OVR

mm - Microsoft Visual C++ - [nbnv.cpp]

File Edit View Insert Project Build Tools Window Help

(Globals) | (All global members) | main

mm classes

```
#include<iostream>
#include<cmath>
#include<cstdlib>
#include<cctype>
using namespace std;
void main()
{

cout<<"toupper(97) : "<<toupper(97)<<endl;
cout<<"toupper(42) : "<<toupper(42)<<endl;
cout<<"sqrt(4):      "<<sqrt(4)<<endl;
cout<<"sqrt(-4):     "<<sqrt(-4)<<endl;
}
```

"C:\Program Files\Microsoft Visual Studi...

```
toupper(97) : 65
toupper(42) : 42
sqrt(4):      2
sqrt(-4):     -1.#IND
Press any key to continue_
```

ClassView | FileView

```
Linking...

mm.exe - 0 error(s), 0 warning(s)
```

Build | Debug | Find in Files 1 | Find in Files 2 | Results | SQL Debugging

Ln 13, Col 2 REC COL OVR READ

start | mm - Microsoft Visual ... | ch4 and ch5 ch6 | Microsoft PowerPoint ... | "C:\Program Files\Mic... | Address | 10:14 AM

```
#include<iostream>
#include<cmath>
#include<cstdlib>
#include<cctype>
using namespace std;
void main()
{

cout<<"toupper(97) : "<<toupper(97)<<endl;
cout<<"toupper(42) : "<<toupper(42)<<endl;
cout<<"sqrt(4):     "<<sqrt(4)<<endl;
cout<<"sqrt(-4):     "<<sqrt(-4)<<endl;
}
```

Char	Value
SP	32
!	33
"	34
#	35
$	36
%	37
&	38
`	39
(	40
)	41
*	42
+	43
,	44
-	45
.	46
/	47
0	48
1	49
2	50
3	51
4	52
5	53
6	54
7	55
8	56
9	57

Char	Value
:	58
;	59
<	60
=	61
>	62
?	63
@	64

[	91
\	92
]	93
^	94
_	95
`	96

{	123
|	124
}	125
~	126
DEL	127

Char	Value
A	65
B	66
C	67
D	68
E	69
F	70
G	71
H	72
I	73
J	74
K	75
L	76
M	77
N	78
O	79
P	80
Q	81
R	82
S	83
T	84
U	85
V	86
W	87
X	88
Y	89
Z	90

Char	Value
a	97
b	98
c	99
d	100
e	101
f	102
g	103
h	104
i	105
j	106
k	107
l	108
m	109
n	110
o	111
p	112
q	113
r	114
s	115
t	116
u	117
v	118
w	119
x	120
y	121
z	122

```
#include<iostream>

using namespace std;
int square( int m );   // function prototype
//it can be int square( int );

int main()
{
   for ( int x = 1; x <= 10; x++ )
      cout << square( x ) << "  ";   //calling statement x is actual parameter

   cout << endl;
   return 0;
}

// Function definition
int square( int y )   // Heading  y is Formal Parameter
{
   return y * y;  // The return Statement
}
```

File Edit View Insert Project Build Tools Window Help

bals) (All global members) square

mm classes

```
#include<iostream>

using namespace std;
int square( int m );    // function prototype

int main()
{
   for ( int x = 1; x <= 10; x++ )
      cout << square( x ) << "  ";//calling statment

   cout << endl;
   return 0;
}

// Function definition
int square( int y )
{
   return y * y;
}
```

"C:\Program Files\Microsoft Visual Studio\My...

```
1  4  9  16  25  36  49  64  81  100
Press any key to continue
```

ClassView FileView

```
inking...

m.exe - 0 error(s), 0 warning(s)
```

Build Debug Find in Files 1 Find in Files 2 Results SQL Debugging

Ln 1, Col 1 REC COL OVR

```
#include<iostream>

using namespace std;
int square( int );   // function prototype

int main()
{
   for ( int x = 1; x <= 10; x++ )
      cout << square( x ) << "  ";

   cout << endl;
   return 0;
}

// Function definition
int square( int y )
{
   return y * y;
}
```

File Edit View Insert Project Build Tools Window Help

bals) | (All global members) | square

mm classes

```
#include<iostream>

using namespace std;
int square( int );   // function prototype

int main()
{
   for ( int x = 1; x <= 10; x++ )
      cout << square( x ) << "  ";

   cout << endl;
   return 0;
}

// Function definition
int square( int y )
{
   return y * y;
}
```

"C:\Program Files\Microsoft Visual Stu...

```
1  4  9  16  25  36  49  64  81  100
Press any key to continue
```

ClassView | FileView

```
inking...

m.exe - 0 error(s), 0 warning(s)
```

Build | Debug | Find in Files 1 | Find in Files 2 | Results | SQL Debugging

Ln 1, Col 1 REC COL OVR

```
#include<iostream>

using namespace std;
int cube( int y ); // function prototype

int main()
{
   int x;

   for ( x = 1; x <= 10; x++ )
      cout << cube( x ) << endl; //calling statement x is actual parameter


   return 0;
}
// Function definition
int cube( int y ) // Heading  y is Formal Parameter

{
   return y * y * y;
}
```

File Edit View Insert Project Build Tools Window Help

bals) | (All global members) | cube

mm classes

```
#include<iostream>

using namespace std;
int cube( int y );

int main()
{
   int x;

   for ( x = 1; x <= 10; x++ )
      cout << cube( x ) << endl;

   return 0;
}

int cube( int y )
{
   return y * y * y;
}
```

"C:\Program Files\Microsoft Visual Studio\My...

```
1
8
27
64
125
216
343
512
729
1000
Press any key to continue_
```

ClassView | FileView

```
inking...

m.exe - 0 error(s), 0 warning(s)
```

Build | Debug | Find in Files 1 | Find in Files 2 | Results | SQL Debugging

Ln 2, Col 1 REC COL OVR

```
#include<iostream>
using namespace std;

int maximum( int, int, int );   // function prototype

int main()
{
   int a, b, c;

   cout << "Enter three integers: "<<endl;
   cin >> a >> b >> c;

   // a, b and c below are arguments (actual parameters) to
   // the maximum function call
   cout << "Maximum is: " << maximum( a, b, c ) << endl;

   return 0;
}

// Function maximum definition
// x, y and z below are parameters ( formal parameters) to
// the maximum function definition
int maximum( int x, int y, int z )
{
   int max = x;

   if ( y > max )
      max = y;
   if ( z > max )
      max = z;
   return max;
}
```

File Edit View Insert Project Build Tools Window Help

bals) (All global members) main

mm classes

```
#include<iostream>

using namespace std;

int maximum( int, int, int );   // function prototype

int main()
{
   int a, b, c;

   cout << "Enter three integers: "<<endl;
   cin >> a >> b >> c;

   // a, b and c below are arguments to
   // the maximum function call
   cout << "Maximum is: " << maximum( a, b, c ) << endl;

   return 0;
}

// Function maximum definition
// x, y and z below are parameters to
// the maximum function definition
int maximum( int x, int y, int z )
{
   int max = x;

   if ( y > max )
      max = y;

   if ( z > max )
      max = z;

   return max;
}
```

"C:\Program Files\Microsoft Visual ...

```
Enter three integers:
100
20
33
Maximum is: 100
Press any key to continue
```

ClassView FileView

inking...

m.exe - 0 error(s), 0 warning(s)

Build Debug Find in Files 1 Find in Files 2 Results SQL Debugging

Ln 11, Col 42 REC COL OVR

Programming Example

Programming Example

```
// Program: Largest
#include <iostream>
using namespace std;
double larger(double x, double y);
int main()
{
   double num; //variable to hold the current number
   double max; //variable to hold the larger number
   int count;  //loop control variable

   cout << "Enter 10 numbers." << endl;
   cin >> num;                          //Step 1
   max = num;                            //Step 1

   for (count = 1; count < 10; count++)       //Step 2
   {
      cin >> num;                       //Step 2a
      max = larger(max, num);                //Step 2b
   }
   cout << "The largest number is " << max<< endl;      //Step 3
   return 0;
}//end main

double larger(double x, double y)
{
   if (x >= y)
      return x;
   else
      return y;
}
```

The `return` Statement

```
#include<iostream>
using namespace std;

int maximum( int, int, int );   // function prototype

int main()
{
   int a, b, c;

   cout << "Enter three integers: "<<endl;
   cin >> a >> b >> c;

   // a, b and c below are arguments (actual parameters) to
   // the maximum function call
   cout << "Maximum is: " << maximum( a, b, c ) << endl;

   return 0;
}

// Function maximum definition
// x, y and z below are parameters ( formal parameters) to
// the maximum function definition
int maximum( int x, int y, int z )
{
   int max = x;

   return 0;

   if ( y > max )
      max = y;
   if ( z > max )
      max = z;

   return max;
}
```

File Edit View Insert Project Build Tools Window Help

bals) | (All global members) | maximum

ggggg classes

```
#include<iostream>
using namespace std;

int maximum( int, int, int );   // function prototype

int main()
{
   int a, b, c;

   cout << "Enter three integers: "<<endl;
   cin >> a >> b >> c;

   // a, b and c below are arguments (actual parameters) to
   // the maximum function call
   cout << "Maximum is: " << maximum( a, b, c ) << endl;

   return 0;
}

// Function maximum definition
// x, y and z below are parameters ( formal parameters) to
// the maximum function definition
int maximum( int x, int y, int z )
{
   int max = x;
 return 0;
   if ( y > max )
      max = y;
   if ( z > max )
      max = z;
   return max;
}
```

"C:\Program Files\Microsoft Vis...

```
Enter three integers:
100
200
52
Maximum is: 0
Press any key to continue_
```

ClassView | FileView

```
inking...

gggg.exe - 0 error(s), 0 warning(s)
```

Build | Debug | Find in Files 1 | Find in Files 2 | Results | SQL Debugging

Ln 26, Col 1 REC COL OVR

```
#include<iostream>
using namespace std;

int maximum( int, int, int );   // function prototype

int main()
{
   int a, b, c;
    return 0;


   cout << "Enter three integers: "<<endl;
   cin >> a >> b >> c;

   // a, b and c below are arguments (actual parameters) to
   // the maximum function call
   cout << "Maximum is: " << maximum( a, b, c ) << endl;

   return 0;
}

// Function maximum definition
// x, y and z below are parameters ( formal parameters) to
// the maximum function definition
int maximum( int x, int y, int z )
{
   int max = x;

   if ( y > max )
      max = y;
   if ( z > max )
      max = z;
   return max;
}
```

File Edit View Insert Project Build Tools Window Help

bals) (All global members) main

ggggg classes

```
#include<iostream>
using namespace std;

int maximum( int, int, int );   // function prototype

int main()
{
   int a, b, c;
 return 0;
   cout << "Enter three integers: "<<endl;
   cin >> a >> b >> c;

   // a, b and c below are arguments (actual paramete
   // the maximum function call
   cout << "Maximum is: " << maximum( a, b, c ) << er

   return 0;
}

// Function maximum definition
// x, y and z below are parameters ( formal parameter
// the maximum function definition
int maximum( int x, int y, int z )
{
   int max = x;

   if ( y > max )
      max = y;
   if ( z > max )
      max = z;
   return max;
}
```

"C:\Program Files\Microsoft Visual Studio\MyP...

```
Press any key to continue_
```

ClassView FileView

```
inking...

gggg.exe - 0 error(s), 0 warning(s)
```

Build Debug Find in Files 1 Find in Files 2 Results SQL Debugging

Ln 9, Col 1 REC COL OVR